

T₃ Uptake

Test Details

METHODOLOGY

Electrochemiluminescence Immunoassay (ECLIA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Thyroid function test for the diagnosis of hypothyroidism or hyperthyroidism, used with thyroxine (T₄) or equivalent to provide free T₄ index, FTI. An indirect measure of binding protein, the T₃ uptake reflects available binding sites (ie, reflects TBG). T₃ uptake is **not** a measurement of serum T₃. It should never be used alone; rather, its usual application is use with thyroxine (T₄).

Special Instructions

This test reflects assessment of thyroxine-binding globulin (TBG) and should not be ordered alone.

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

Alterations in binding capacity of TBG are described with major illness and with high doses of salicylates and corticosteroids; with use of heroin, methadone, phenytoin, and perphenazine. Alterations occur with malnutrition, such as in metastatic malignancy, and are found in patients with abnormal serum protein patterns (eg, nephrotic syndromes, cirrhosis). Other states in which changes in TBG occur include infancy, acromegaly, molar and ordinary pregnancy, oral contraceptives, and with exogenous hormones including androgens, anabolic steroids, and estrogens. Hereditary increase and decrease of TBG occurs.

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL (adult), 0.8 mL (pediatric)

Minimum Volume

0.7 mL (adult), 0.3 mL (pediatric) (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

If a red-top tube or plasma is used, transfer separated serum or plasma to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.¹

Age	Male (%)	Female (%)
0 to 11 m	23–34	23–36
1 to 3 y	24–35	24–36
4 to 6 y	24–34	24–35
7 to 11 y	24–33	22–35
12 to 15 y	25–37	23–37
16 to 18 y	24–38	23–35
>18 y	24–39	24–39

Storage Instructions

Room temperature

References

Bakerman S. *A, B, C's of Interpretive Laboratory Data*. Greenville, NC: Interpretive Laboratory Data Inc;1984.

Burke MD. Thyroid function studies. Test strategies and interpretation of results. *Postgrad Med*. 1980 Dec; 68(6):169-177. PubMed 6776511

(<http://www.ncbi.nlm.nih.gov/pubmed/6776511>)

Gruhn JG, Barsano CP, Kumar Y. The development of tests of thyroid function. *Arch Pathol Lab Med*. 1987 Jan; 111(1):84-100. PubMed 3541847

(<http://www.ncbi.nlm.nih.gov/pubmed/3541847>)

Footnotes

1. Soldin SJ, Cook J, Beatey J, et al. Pediatric reference ranges for thyroxine and tri-iodothyronine uptake. *Clin Chem*. 1992; 38:960.